The ML System Design Framework

Scaler School of Technology | ML System Design and LLMOps | Week 1 — Session 2/2 | Pre-Lecture Prep

What Is This Lecture About?

This session moves from "how ML systems are built" to "why they are built the way they are." We will ground every architectural decision in Chip Huyen's framework from *Designing ML Systems* (O'Reilly, 2022), Chapter 2. You will leave with a repeatable process for scoping, framing, and designing any ML system from a blank page.

Business Objectives First

Most ML projects fail not because the model is bad, but because the team optimised the wrong metric. We will learn to trace a clear path from a business KPI to an ML objective function — using Netflix's take-rate as a worked example.

Four Requirements That Drive Architecture

Every ML system must be reliable, scalable, maintainable, and adaptable. We will see exactly what each requirement means in practice: silent failures, 8,000-model deployments, and systems that adapt to shifting data without going down.

Iterative Process and Problem Framing

ML development is never a straight line. We will walk the 6-step cycle (scope → data → model → deploy → monitor → analyse → repeat) and learn how problem framing — the choice of inputs, outputs, and objective function — can make a system easy to maintain or impossible to scale.

What to Expect in the Session

- Opening quiz: what business metric does your favourite ML use case move?
- The business-to-ML metric translation table
- Netflix take-rate example and causal chain walk-through
- The four requirements and their failure modes
- The 6-step iterative development cycle diagram
- Problem framing exercise: from a business problem to input/output/objective
- Decoupling objectives: live example with newsfeed ranking
- Mind vs. data debate: Norvig, Sutton, and Pearl

How to Prepare

- Think of an ML project you would like to build — real or hypothetical. What **business metric** would it move? How would you know if it worked?
- Recall the 5-question framework from the last session (Problem → Data → Model → Serving → Monitoring). This session goes deep on Question 1: the problem.
- Optional: skim Chapter 2 of *Designing ML Systems* (Chip Huyen, O'Reilly 2022). Focus on the "Framing ML Problems" section.
- Optional: read Rich Sutton's "The Bitter Lesson" (2019) — freely available at `incompleteideas.net`. It is a 5-minute read that will make the mind vs. data discussion much richer.
- No new tools or setup needed today.

Duration: ∼2 hours **Source:** Chip Huyen, *Designing ML Systems*, Ch. 2 (O'Reilly, 2022) **Instructor:** Priyansh Saxena